

PORTFOLIO

YOUSRA IGOUZAL

INCLUDES TECHNICAL PROJECTS AND LEADERSHIP EXPERIENCE

YOUSRA.IGOUZAL@GMAIL.COM
+44 7557 561494

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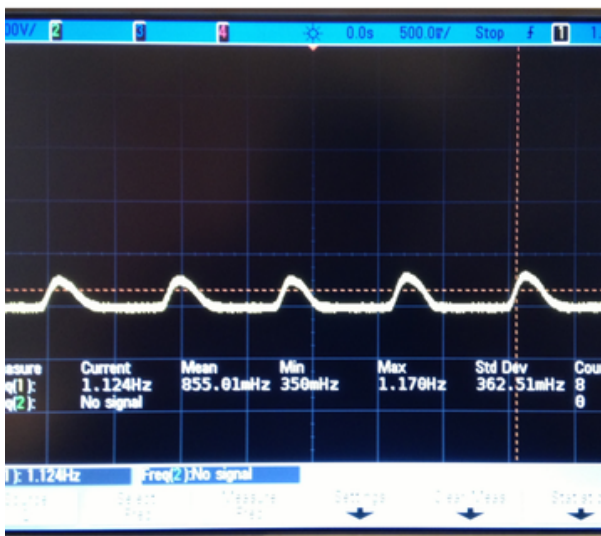
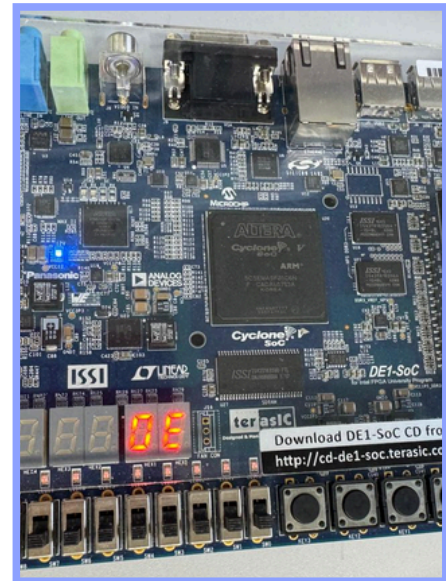
TECHNICAL PROJECTS

PROJECT 1 : FPGA MICROPROCESSOR INSTRUCTION SET EXTENSION

Tools: DE1-SoC FPGA, SystemVerilog, Quartus, ModelSim

Achievement:

1. Designed and integrated memory mapped inputs and outputs to interface physical switches and 7 segment displays direct via the processor's data bus.
2. Successfully expanded a basic microprocessor's instruction set by adding a custom XOR opcode to enable encryption and decryption algorithms.



PROJECT 2 : HEART RATE MONITOR USING OPTICAL SENSOR

Tools: AVR, optical sensor, oscillogram, ADC

Achievement:

Built a functional photoplethysmography system to detect heart rate from a fingertip.

Implementation:

Sampled an infrared sensor and custom filter circuit with the ADC, detected beats in firmware, and displayed BPM using a UART and LED feedback.

PROJECT 3 : INTERRUPT-DRIVEN SYSTEM DESIGN

Tools: AVR, Logic Analyzer

Achievement:

Converted polling based systems to interrupt driven architecture in order to improve CPU's efficiency.

Implementation:

Handled timer interrupts for periodic tasks, ADC interrupts for sensor data, and external interrupts for button presses. Measured and optimized ISR execution time.

PROJECT 4 : SERIAL COMMUNICATION & HARDWARE DEBUG INTERFACES

Tools: AVR ATmega644P, SPI, I²C, UART, Logic Analyzer, MCP4261, MCP7940M

Achievement:

Established full communication with three different peripheral types using UART to debug, SPI as a digital pot, and I²C for the real time clock.

Implementation:

Developed drivers to get potentiometer values and read and write registers. Created a UART debug console for real-time system monitoring and data logging.

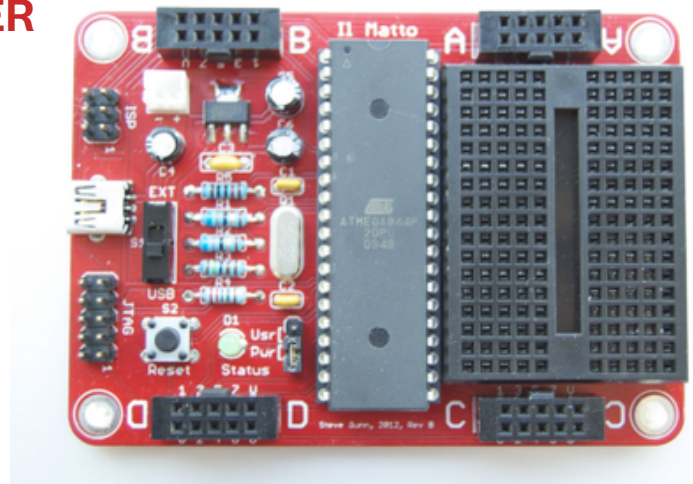
PROJECT 5 : MICROCONTROLLER ASSEMBLY

Tools : Soldering Iron, Iron stand, Solder, Desoldering tool, Blue-tac

Components : Il Matto Kit, C232HM cable, Hook-up wire

Equipment : Oscilloscope, Logic Analyser, Bench PSU, Multimeter

Software : AVRDUDE, avr-gcc, UrJTAC



Assembled, and tested a custom microcontroller development board based on the Atmel AVR microcontroller. The project involved building the complete system from component assembly (soldering) through functional verification, applying industry-standard PCB assembly and testing methodologies. This board was subsequently used during laboratory sessions to gain experience with serial communications, timers, counters, PWM, and analogue inputs and other C programming principles.

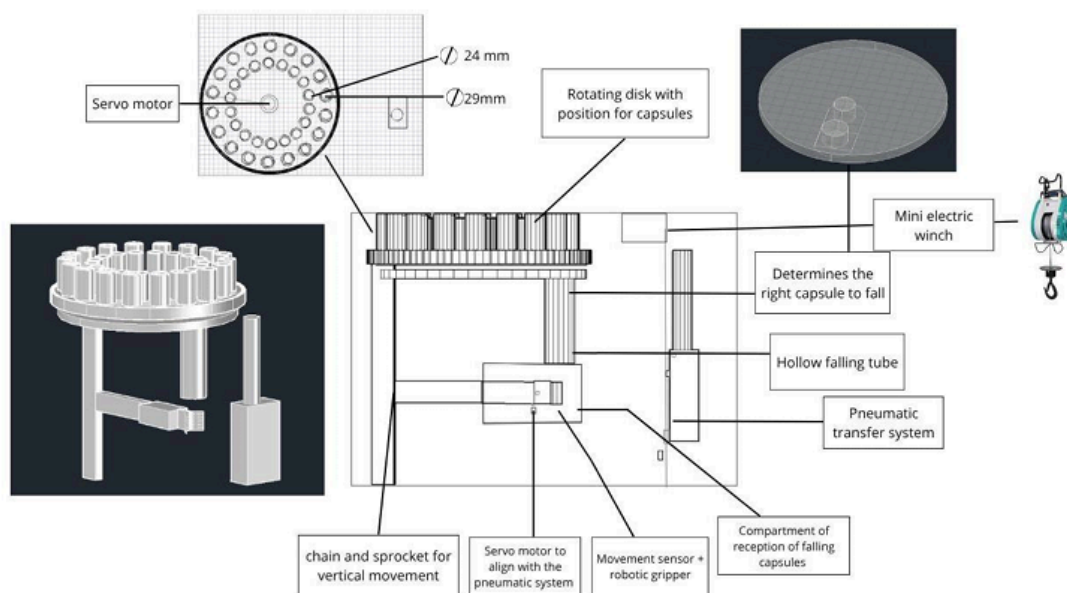
PROJECT 6 : BATTERY FOR SOLAR UAV

With the **Renewable and Sustainable Engineering society** at the University of Southampton (RESUS), I am part of the electronics department, primarily responsible for the battery subsystem. So far, I have been in charge of researching and deciding the optimal battery type and configuration to use and organising soldering sessions to build the custom battery type. The next phase involves collaborating with the solar team to integrate solar panel output into the battery system.



PROJECT 7: SAMPLE HANDLER

From June to August 2025, I interned at the National Center for Nuclear Energy in Morocco. My role was to automate a technique used at the center: Neutronic Activation Analysis. One of the steps was manual and tedious in the lab, therefore, I prototyped a sample handler, in accordance with the architecture of the Laboratory, the materials to use in nuclear exposed settings to automate the process. The sample handler improves efficiency by allowing 300 additional samples per year. During the project, I applied engineering research methodologies, used AutoCAD, and produced technical and assembly drawings and constantly communicated with the engineers about ways to improve the prototype. A final report on the prototype was submitted and is currently under review by the Quality and Safety Department. If approved, it will be implemented in the lab.



TECHNICAL SKILLS SUMMARY

Hardware: FPGA | Oscilloscopes | Logic Analyzers | Multimeters | Soldering

Software: Quartus | ModelSim | AutoCAD | SystemVerilog | C | Python | MATLAB

LEADERSHIP PROJECTS



PETSSS - SOCIAL ENTREPRISE

I am project lead and Founder of Petsss, an ENACTUS social enterprise rescuing and training stray dogs and cats in Morocco.

Being lead has allowed me to develop huge commercial awareness as we were looking for a way to fund our operations. We conducted market research by asking dog owners their needs. We found a market gap for waste bags and for hypoallergenic soap in Southampton.

Operations were led in 2 countries, in Southampton with a team of 20 students. In Morocco with 3 individuals. And with 3 pet shelters in Morocco.

We directly rescued 7 dogs and indirectly helped 3 pet shelters by sending our team to Morocco.

Petsss won 2nd place out of 60 projects at ENACTUS UK.

VICE PRESIDENT OF ENGAGEMENT

I am also Vice president of Engagement of the Renewability and Sustainability Engineering Society. I lead member recruitment during freshers week and worked with the Engineering magazine to display the work done at resus. I am also constantly looking to collaborate with more societies and engineering organisation to increase engagement withing the society.



SOFT SKILLS SUMMARY

Project management | Team coordination | Stakeholder engagement | Public speaking | Member recruitment | Technical presentations | Multilingual (English, French, Arabic, Spanish)| Initiative | self-motivation | Problem-solving | Decision-making | Resilience